

3D Printing and Modeling

1st lab: Introduction to our laboratory and basics of Fusion 360

0. Resources

1. [How To Use Revolve Feature...](#) (2-3 min)
2. [Getting Started with the SWEEP TOOL](#) (11 min)
3. [Fusion 360 User Interface Tutorial](#) (5 min)
4. [How to create a rectangle in Fusion 360](#)
5. [Getting Started with Materials](#) (12-13 min)

1. The Mug



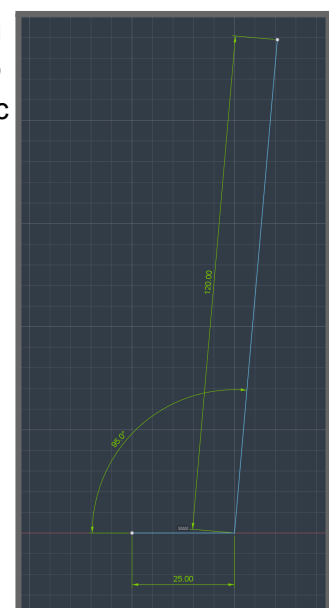
In this laboratory, we will start by learning to create a mug similar to this one, by using a couple new tools: **Revolve & Sweep**

1.1 Initial design

We will start our design by creating what is essentially going to be the vertical section (*the view that we would have if we were to cut the mug in half*) of our object. This view is going to be symmetric in relation to a central axis. Therefore, we will only be sketching a half.

Create two lines at an angle of around 95° , just like the image on the right. You can use the **Sketch Dimension** tool to specify this angle. These are going to be the exterior walls of our mug.

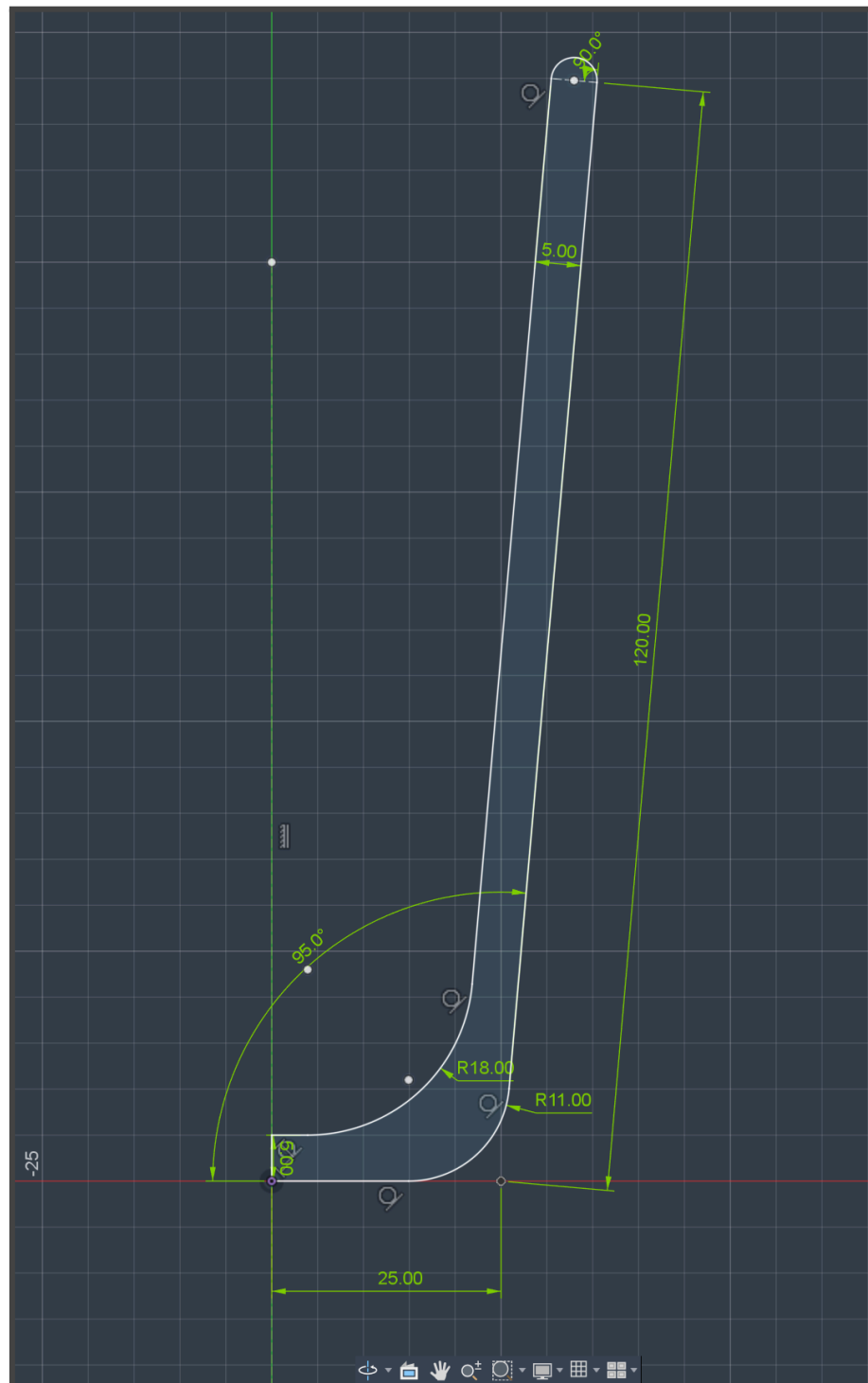
Continue the sketch by creating the interior walls and connecting them with the exterior walls. For this example, I've created a circle between the two points at the top and filleted the corners.



I recommend using **Offset** to start the interior walls.

You can fillet the corners by using the **Fillet** tool from the top menu. It works similarly to the tool of the same name in the direct modeling menu.

In the end, the sketch should look something like this:

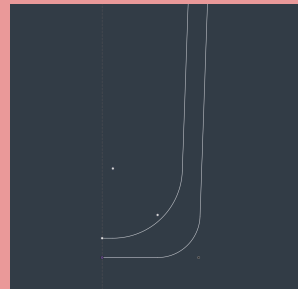


1.2. Revolve

Revolve helps us turn a 2D profile into a 3D body along a curved or straight path, not just perpendicular to a plane like **Extrude**. It is used to model shapes such as pipes, cables, rails, or any part that follows a curve in space.

You can access this tool from the toolbar, next to **Extrude**.

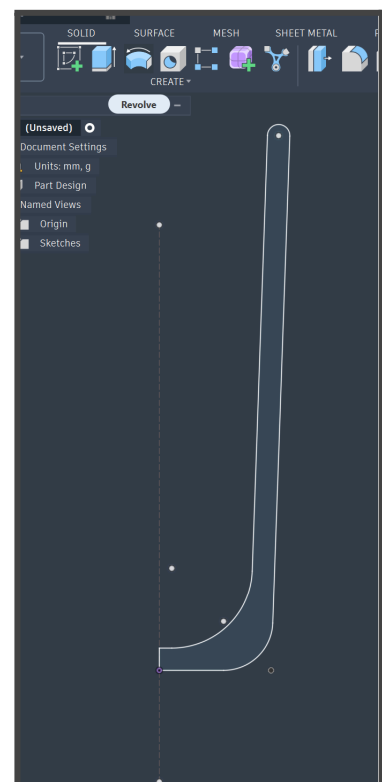
Make sure the shape is closed before trying to select it.
The image is a **bad** example.



Revolve works by rotating the sketch around a given axis. Therefore, the sketch needs to also contain this axis. You can add it manually as a construction line. The image on the right is a good example of how it is supposed to look.

Click **Revolve** and then select both the area you wish to rotate and the axis of rotation. Just like **Extrude**, you can select the direction of rotation, the range in degrees and operation type. For our mug, we will revolve the full 360 degrees.

The result should look something like this:



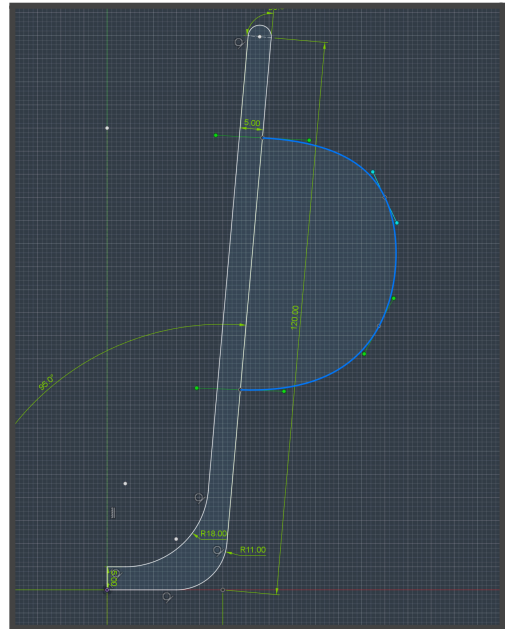
1.3. The handle

We will construct the handle by dragging a 2D profile along a path. For this, we will need to learn about **Sweep**. It works by taking a sketch (profile) and stretching it along a specified path. Again, unlike extrude which extrudes only in one direction, sweep can follow any path in our design.

Start by creating the path you wish the handle to follow. I suggest using the **Fit Point Spline** tool (Create > Fit Point Spline) from the sketching toolbar to create the curved line.

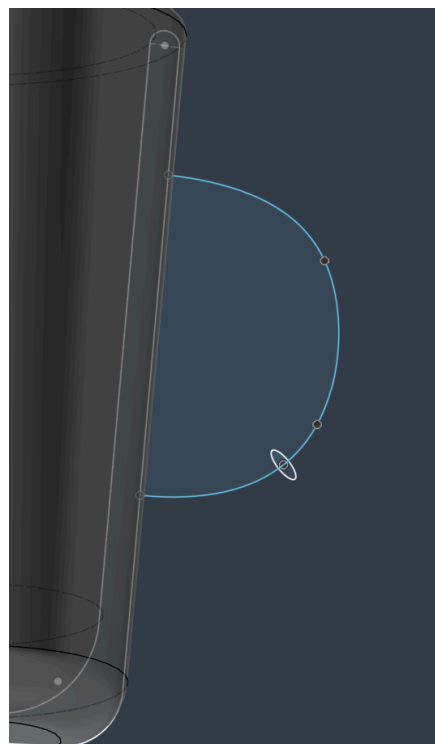
Once you are happy with your path, you will need to create the shape of the handle. For that we will need a new sketch that is perpendicular to our path at any point.

Construct > **Plane Along Path** and click any point on your path. Now create a new sketch on that plane. This new sketch will act as the section of the handle. A simple circle will be enough.



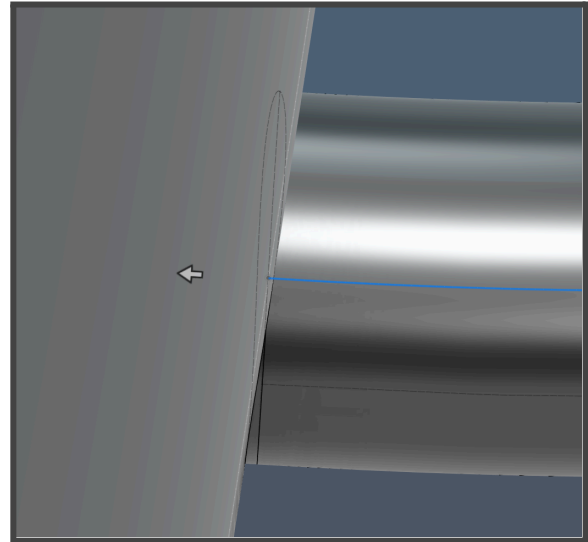
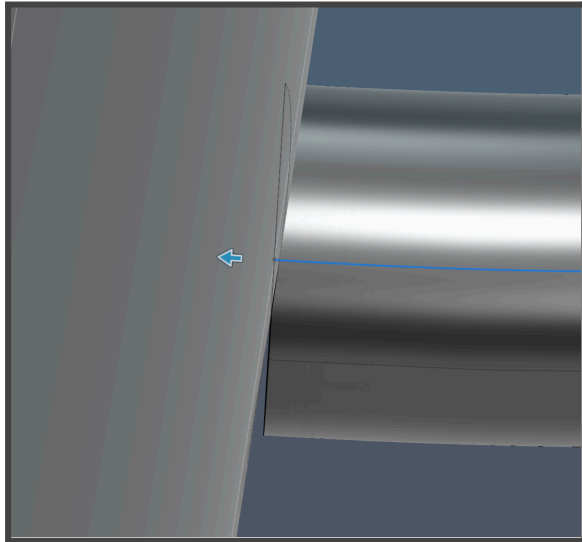
When creating a **Plane Along Path**, the new plane will be centered (intersection of the X and Y axis) on the path. For our handle to follow the path exactly it needs to also be centered in the origin point.

After these steps, our design should look something like this:



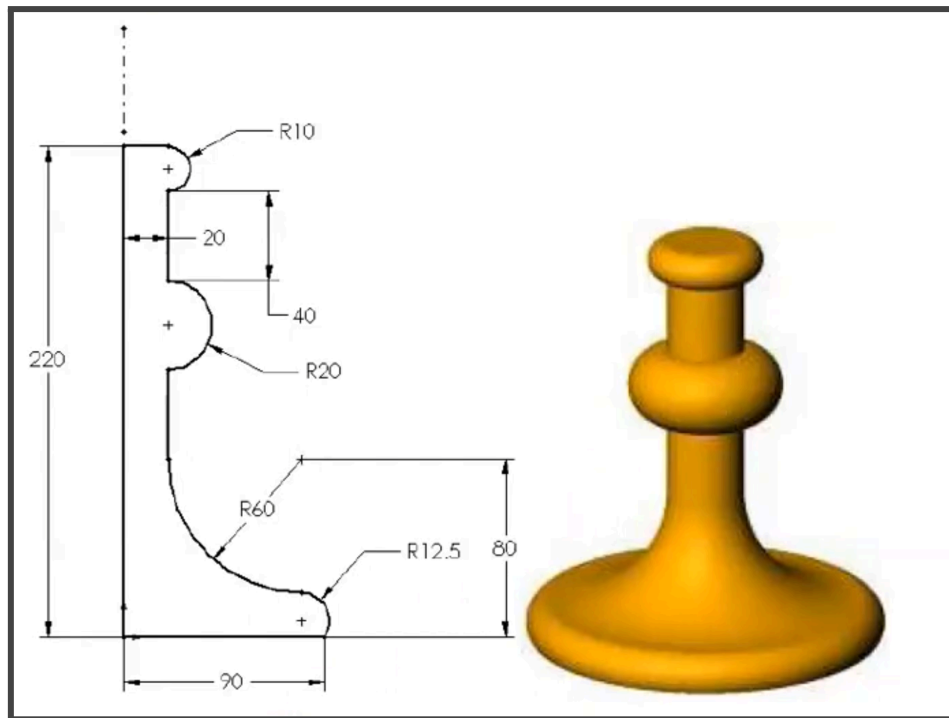
1.3.1. *Sweep*

We are now ready for the final step. Go to **Create > Sweep**. Select the profile and the path it will follow. If you created the plane in the middle of the path, you will need to extrude on both sides of it. Adjust the distances until satisfied. Make sure the handle joins the body of the mug.

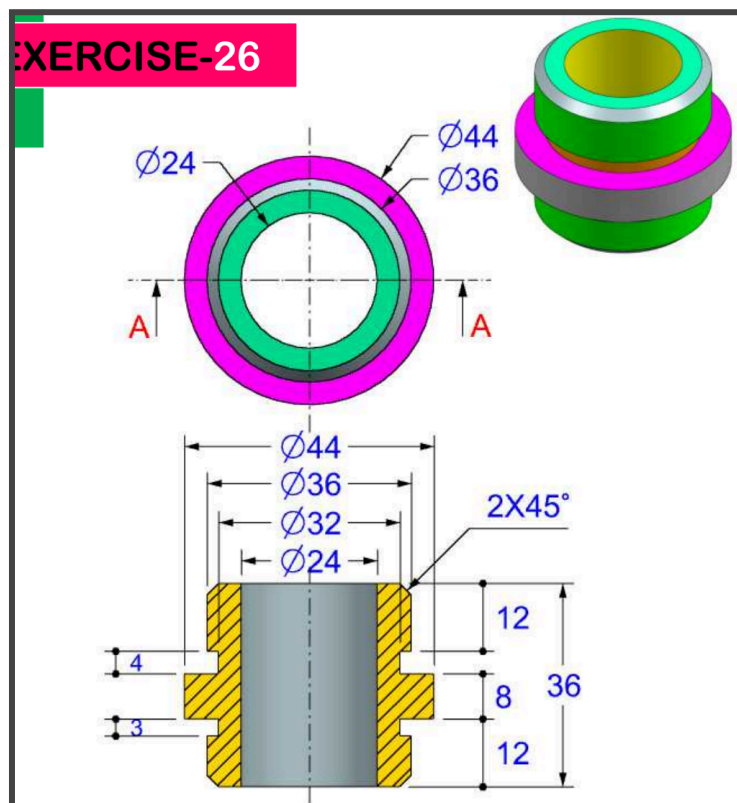


2. Exercises

- Exercise 1: Model a perfect sphere of any dimensions.
- Exercise 2:



- Exercise 3:



4. Exercise 4 (hard):

